University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,852: User-Centred Design

ECTS credits: 6

Overview examination/s

(binding regulations see below)
central - Written examination, Analog, Individual work individual grade (60%, 120 mins.)
Examination time: Lecture-free period
decentral - Portfolio, --, -- (40%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,852,1.00 User-Centred Design -- English -- Schöning Johannes
7,852,2.01 User-Centred Design: Exercises, Group 1 -- English -- Kaltenhauser Annika
7,852,2.02 User-Centred Design: Exercises, Group 2 -- English -- Preussner Adrian

Course information

Prerequisites

There are no required prerequisites. The course is open for **both MCS and MBI students.**

Learning objectives

- Knowledge of core concepts of human-computer interaction
- Knowledge of motor, perceptive and cognitive abilities of humans
- Knowledge of interaction paradigms
- Knowledge of current interaction technologies
- Competences in user-centred design and evaluation
- Ability to design simple user interfaces with high usability
- Ability to design simple user interfaces with great user experiences
- Ability to conduct user interface evaluations

Content

In this lecture, we focus on user-centred design as a core principle of human-computer interaction (HCI). We examine how an understanding of human cognition, ergonomics, and human information processing informs the design of interactive systems. Special attention is given to ergonomic and physiological principles, such as motor functions and visual perception, to ensure that systems support users effectively.

The lecture introduces fundamental interaction paradigms, devices, and interaction styles while maintaining a consistent emphasis on how these technologies can be shaped to serve the user. A structured, user-centred design process forms the foundation of the lecture content. This includes methods such as the development of scenarios, iterative prototyping, and systematic evaluation.

Participants learn how to apply user-centred approaches to all levels of system design, including activity-level considerations, information and interaction design, and technical implementation. The objective is to enable the creation of interactive systems with high usability and excellent user experience, grounded in a clear understanding of users and their context.

Structure and indications of the learning and teaching design

The course consists of weekly sessions (2-hour lecture and 2-hour exercise). The content of the exercises will closely build upon the lecture content. Lecture notes via the provided slides and exercises. The lectures will cover the following topics:

- Introduction, Basic Terms & Definitions
- History and Paradigms of HCI
- Interaction Design Process
- Ideation Methods & Prototyping
- Sketching Paper & Figma
- Design Rules and HCI Principles
- UI Evaluation
- Analysis of log data
- Dark Patterns, Ethical Challenges & Accessibility
- Models of Interaction: 7 Stages of Action
- Fitts Law
- HCI & AI

Literature

- Rogers, Yvonne, Helen Sharp, and Jenny Preece. Interaction design: beyond human-computer interaction. John Wiley & Sons, 2011.
- Dix, Alan, et al. Human-computer interaction. Pearson Education, 2003.
- Shneiderman, Ben, et al. Designing the user interface: strategies for effective human-computer interaction. Pearson, 2016.
- Norman, Don. The Design of everyday things: Revised and expanded edition. 2013.

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Written examination
Responsible for organisation	central
Examination form	Written exam
Examination mode	Analog
Time of examination	Lecture-free period
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	60%
Duration	120 mins.

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are admissible must be explicitly listed by faculty members in the paragraph entitled “Supplementary aids” of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

None

2. Examination sub part (2/2)

Examination modalities

Examination type	Portfolio
Responsible for organisation	decentral
Examination form	Mixed form
Examination mode	--
Time of examination	Term time
Examination execution	--
Examination location	--
Grading type	--
Weighting	40%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

no regulation necessary

- For written examinations at home (term paper), courses without credits, etc., no specific rules for examination aids are required.
- The regulations of the University of St. Gallen and the rules of academic work (sources and aids must always be identified) are applicable in a subsidiary fashion.
- All written work must be accompanied by a declaration of authorship.

Supplementary aids

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Content

Group Exercises (Portfolio)
Students will conduct six activities over the course of the semester. As part of these exercises, the students will apply human-centred design principles to create multiple prototypes for the interface of a software product. The result of the group work will be presented at the end of the semester.

- Assignment 1: Pitch it!
- Assignment 2: Sketching + Paper Prototyping
- Assignment 3: Figma
- Assignment 4: Study Design
- Assignment 5: Study Evaluation
- Assignment 6: Presentation (in class)

The slides will be uploaded weekly for each of the sessions. All materials relevant to the final written examination will be available at the latest after the last lecture.

Literature

1. Lecture notes via the provided slides and exercises.
2. Rogers, Yvonne, Helen Sharp, and Jenny Preece. Interaction design: beyond human-computer interaction. John Wiley & Sons, 2011. (Required)
3. Norman, Don. The Design of everyday things: Revised and expanded edition. 2013. (Required)
4. Dix, Alan, et al. Human-computer interaction. Pearson Education, 2003. (Supplementary)
5. Shneiderman, Ben, et al. Designing the user interface: strategies for effective human-computer interaction. Pearson, 2016. (Supplementary)

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).